

Insight Summary For Stakeholders

**Dataset Used: Adult Psychiatric Morbidity
Survey (APMS) 2014**

Project Name: AI MENTAL HEALTH

Submitted by –

Deepak Baranwal

Data Analytics Intern

DATE: 09 May,2026

Submitted to –

Ms. Jasmine

Team Leader

Zeno Talent

Executive Summary:

This report presents a summary of the mental health data analysis performed using the Adult Psychiatric Morbidity Survey (APMS) 2014 dataset provided by NHS Digital. The analysis focused on identifying demographic and regional patterns in Common Mental Disorders (CMD) across England.

The demographic analysis showed noticeable differences in mental health prevalence across gender and severity categories. Female participants generally showed slightly higher prevalence percentages in higher severity categories compared to males.

Regional analysis also revealed variation in mental health disorder prevalence across different regions of England. The South West region showed comparatively higher CMD prevalence, while the South East region showed lower prevalence levels.

Different visualizations such as grouped bar charts and regional intensity maps were created to improve understanding of the data and support easier interpretation of regional patterns.

Overall, the analysis highlights the importance of demographic and regional mental health monitoring and demonstrates how data visualization can support public health decision-making.

Introduction:

Mental health disorders are an important public health concern in the United Kingdom. Understanding how mental health prevalence differs across demographic groups and geographical regions can help healthcare organizations improve planning, awareness, and support services.

This analysis was performed using the Adult Psychiatric Morbidity Survey (APMS) 2014 dataset published by NHS Digital. The objective of the analysis was to study demographic and regional patterns related to Common Mental Disorders (CMD) across England.

The work completed during Week 1 included dataset profiling, demographic analysis, and regional prevalence mapping. Different visualization techniques such as bar charts and regional intensity maps were used to better understand the distribution of mental health disorders across populations and regions.

The findings presented in this report are intended to provide a simple and understandable summary for stakeholders and non-technical audiences.

Dataset Overview:

The dataset used in this analysis was taken from the Adult Psychiatric Morbidity Survey (APMS) 2014 published by NHS Digital. The dataset contains statistical information related to mental health conditions among adults living in England.

The analysis focused on Common Mental Disorders (CMD) and included demographic as well as regional prevalence data. Different sections of the dataset were used to analyse gender-based differences and regional variation in mental health disorder prevalence.

The dataset included information related to:

- Mental health severity categories
- Gender-based prevalence values
- Regional prevalence percentages
- Age-standardised CMD prevalence data

The extracted data was processed and visualized using Python libraries such as Pandas, Matplotlib, and GeoPandas.

	Men	Women	All Adults
Category			
0-5	74.7125	64.0875	69.2750
6-11	13.7750	17.2375	15.5375
Under 12	88.4500	81.3125	84.8125
12-17	5.6000	9.1625	7.3875
18 or more	5.9375	9.5250	7.7625
12 or more	11.5500	18.6875	15.1875

...	Region	Any_CMD
0	North East	15.7
1	North West	19.0
2	Yorkshire and The Humber	16.7
3	East Midlands	16.7
4	West Midlands	18.4
5	East of England	14.4
6	London	18.0
7	South East	13.6
8	South West	20.9

Key Findings from ANA-1 (Dataset Profiling):

The first stage of the analysis focused on understanding the structure and contents of the APMS 2014 dataset. Different tables related to Common Mental Disorders (CMD) were explored to identify useful variables for demographic and regional analysis.

The dataset contained prevalence percentages categorized by severity levels, gender groups, and geographical regions. During the profiling stage, relevant sections of the dataset were identified and selected for further analysis.

Basic preprocessing steps such as extracting required rows, organizing categories, and preparing clean dataframes were performed before visualization and analysis.

This initial profiling stage helped in understanding the dataset structure and selecting suitable data for demographic comparison and regional prevalence mapping.

Key Findings from ANA-2 (Demographic Analysis):

The demographic analysis focused on comparing Common Mental Disorder (CMD) prevalence across different gender groups using the APMS 2014 dataset.

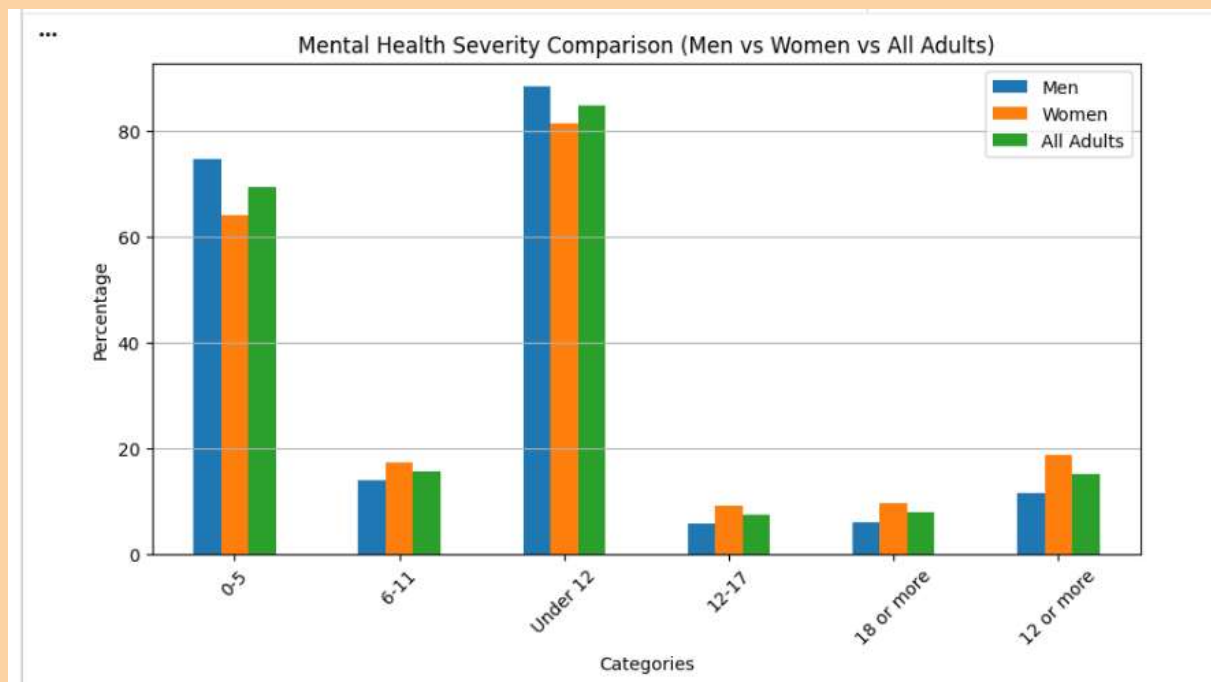
Mental health severity categories such as low, moderate, and severe prevalence levels were compared between men, women, and overall adult populations. The analysis showed noticeable variation between gender groups across different severity categories.

Women generally showed slightly higher prevalence percentages in moderate and severe mental health categories, while men showed comparatively higher values in lower severity categories.

Grouped bar charts were created to visually compare prevalence differences across demographic groups. These visualizations helped in identifying mental health patterns more clearly and made the comparison easier to understand.

Overall, the demographic analysis highlighted the importance of studying mental health prevalence across different population groups.

...	Men	Women	All Adults
Category			
0-5	74.7125	64.0875	69.2750
6-11	13.7750	17.2375	15.5375
Under 12	88.4500	81.3125	84.8125
12-17	5.6000	9.1625	7.3875
18 or more	5.9375	9.5250	7.7625
12 or more	11.5500	18.6875	15.1875



Key Findings from ANA-3 (Regional Analysis):

The regional analysis focused on identifying variation in Common Mental Disorder (CMD) prevalence across different regions of England using age-standardised prevalence values from the APMS 2014 dataset.

Different regions showed noticeable differences in mental health disorder prevalence. The South West region showed comparatively higher prevalence rates, while the South East region showed lower prevalence levels.

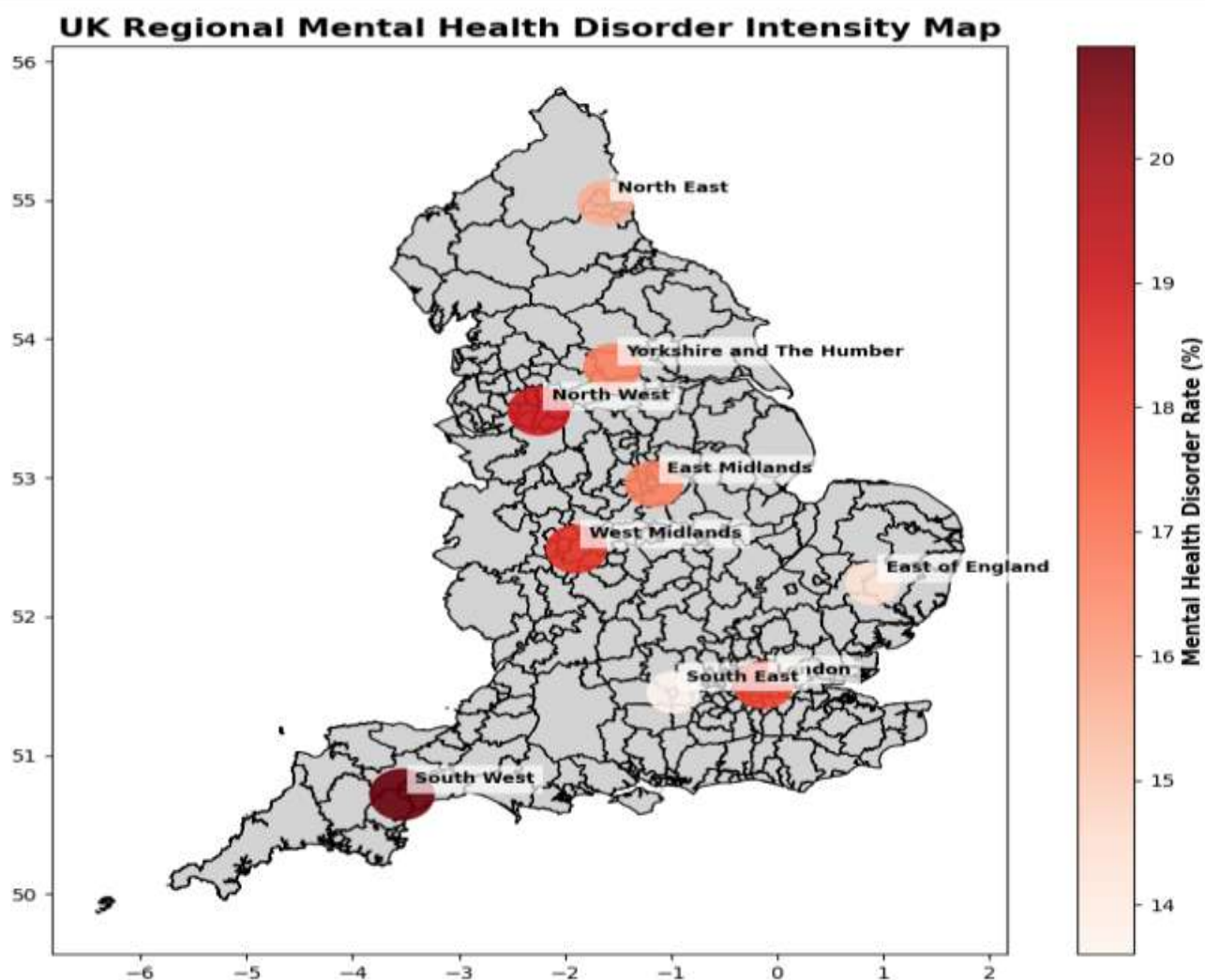
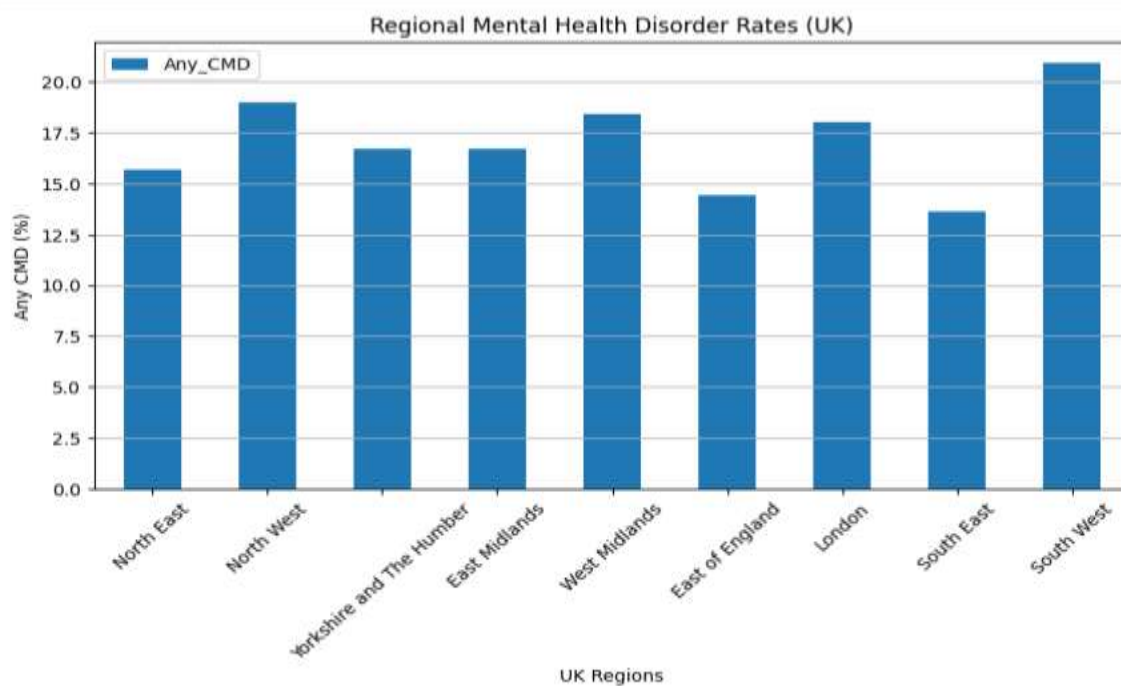
Regional bar charts and mental health intensity maps were created to visualize prevalence variation across regions. The intensity map used color scaling and proportional markers to represent prevalence differences more clearly.

Regions with darker colors and larger markers represented higher prevalence values, while lighter colors and smaller markers represented lower prevalence levels.

The regional analysis helped in understanding geographical differences in mental health prevalence and demonstrated how visualization techniques can support public health analysis and decision-making.

	Region	Any_CMD
0	North East	15.7
1	North West	19.0
2	Yorkshire and The Humber	16.7
3	East Midlands	16.7
4	West Midlands	18.4
5	East of England	14.4
6	London	18.0
7	South East	13.6
8	South West	20.9

Next steps: [Generate code with region_data](#) [New interactive sheet](#)



Visual Highlights:

Different visualization techniques were used throughout the analysis to better understand demographic and regional mental health patterns across England.

Grouped bar charts were used to compare prevalence values across gender groups, while regional bar charts and intensity maps were used to study geographical variation in Common Mental Disorder prevalence.

These visualizations helped simplify the interpretation of the dataset and made the findings easier to understand for non-technical audiences and stakeholders.

Clinical Insights:

The analysis indicates that mental health prevalence varies across demographic groups and geographical regions. Female participants showed slightly higher prevalence levels in moderate and severe mental health categories compared to males.

Regional analysis also highlighted that some regions showed comparatively higher Common Mental Disorder prevalence than others. Such differences may be associated with social, economic, lifestyle, and healthcare-related factors.

These findings emphasize the importance of early mental health awareness, regional healthcare support, and continued public health monitoring for improving mental health outcomes.

Conclusion:

This analysis successfully explored demographic and regional patterns related to Common Mental Disorders (CMD) using the APMS 2014 dataset published by NHS Digital.

The demographic analysis showed variation in mental health prevalence across different gender groups and severity categories. The regional analysis also identified noticeable differences in CMD prevalence across various regions of England.

Different visualization techniques such as grouped bar charts, regional comparison charts, and intensity maps helped in understanding the dataset more clearly and simplified the interpretation of mental health patterns.

Overall, the analysis demonstrated how data visualization and regional analysis techniques can support public health understanding, healthcare planning, and stakeholder decision-making.

The findings from this study can help support future mental health awareness initiatives and encourage further analysis of mental health trends across different populations and regions.